

PROCESS FOLLOWED TO DO THE 2021 RESIDENTIAL BILLING –

Last update: 9-15-21

GENERAL NOTES:

1. The primary initial data sources are:
 - a. Master Sewer Database – which originated as a MSACCESS database but was ported to SQL Server by consultant Tim Curtin. I do an ODBC link to that SQL database to pull data for sewer connection status as well as connection type (residential vs commercial). See below for more specific details
 - b. CWC Data – The Connecticut Water Company provides “annual” data to us in January for all accounts in Avon – commercial and residential. Their data is not particularly “clean” and requires a fair amount of work to make it useable. Tim Foster uses this data to determine the Commercial billing. The reason I say “annual” is because for residential accounts, they read meters quarterly. Avon is essentially split into 3 meter reading zones, and they stagger the readings for each zone by one month. This means that for some residents, the annual amount used is January to December, while others is February to January or March to February. My understanding is that non-residential accounts are read monthly. The data also has inconsistencies with how they identify things like primary vs deduct (irrigation) meters.
 - c. PARCEL.TXT – an extract from the Assessor’s database run nightly. This data source provides current ownership information such as name and primary address, as well as the Avon parcel address. It also has water supply information and sales history

PROCESS:

1. Step 1 was to get all of the data needed together and in a useable form. The data files used are:
 - a. N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\CT Water data\Calendar Year 2020\Avondv123120_original.txt – this is the original CWC text file for residential consumption. It was sent by Joann Detoro on January 4, 2021
 - b. N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\CT Water data\Calendar Year 2020\Avondv123120.txt – this is the original CWC text file for residential consumption after being modified to facilitate connection to Avon data. (See d below)
 - c. Text file from Jim Grimaldi N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\SEWER USE CHARGES\USE CHARGE-2021\ 9-13-21 from JG FOR Oct2021.xlsx – this is used to get the LIST NUMBER and delinquent accounts from the Collector of revenue – using the parcel number as the link. In fact, what I did was to prepare the linking files, develop the costs/residence, and submit this information as strictly the ParcelID and cost as two columns in a text file, and submit that to Deb Fioretti → Jim Grimaldi. They prepared the file that includes the parcelID, List number and several other data items which I use later on within Access.
 - d. N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\SEWER USE CHARGES\CW_TOA_LUT\ CWctoTOA_lookuptable.xls – this is a look-up-table that I prepared in which I link the parcelID to the CWC account number (specifically called the serviceaddr) – I create a new field called servacctno stripping the word **Avon** from the data. Joann Detoro from CWC told me that this number does not change with ownership but stays with the address, so it should be good from year to year regardless of owner. In the future, this file will be critical to maintain with new accounts as part of the initial process to billing as it becomes the key to

determining the October sewer billing for residential accounts. We also don't know when CWC adds new accounts in Avon

- e. N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\SEWER USE CHARGES\USE CHARGE-2021\FINAL 2021 residential on CWC October .xlsx – this excel file has several worksheets in it:
- Use this table for billing** is the most important table in it as it contains processed data from the master sewer database, CWC, identifies the low 2 quarters of water use, does the math for calculating the charges, and has much of the data to be used later during the mail merge to produce the bills. Some of this data came from an output of queries from MS Access. Ultimately, I added well user data below the public water user data so that this could become a source for checking and links within MS Access. The primary value of using EXCEL was the ability to identify the low two quarters, double them, and determine the rate to be used / 1000 gallons of treatment. That was an iterative process until a revenue target was achieved.
 - Well accounts** – this is an output from a query in MS Access in which the well accounts are identified and the cost per is summed. Well-based vs public water users is included within the master sewer database as well as the Parcel.txt file (CAMA), however this must be checked annually since we don't know when CWC adds new accounts in Avon.
 - 2021 residential on CWC_Well OC** – this table contains the Parcel ID and October cost for each account can be deleted
 - Use this table for Grimaldi** – this table contains the Parcel ID and October cost for each account and was used to create the text file that was sent to Deb F/Jim Grimaldi so that they could add their billing number to it.
 - Old tables that were trial and error include:
 - CWC to LUT to Swrmaster 1-11-21
 - Old
- f. N:\ENGINEERING_SEWER_FILES\SEWER BILLING & CHARGES\SEWER USE CHARGES\USE CHARGE-2021\sewer master linked to sql and CWC for sewer billing.accdb – this is the MS Access database containing many tables and queries:
- Link to the master sewer SQL database – specifically table **dbo_MASTER_SEWER_DATABASE**. This is our main sewer database table and critical to the management of sewers and sewer billing
 - Imported **PARCEL.TXT** – there are several parcel txt files in the database. The last one used was dated 9-13-21. I originally had this as a linked table rather than one loaded into the database, thinking that as a linked table, every time I opened the database I would get the most up-to-date data. Unfortunately, as a linked table, I can't change the Parcel ID from number to text or vice-versa which means I can't do queries linking all of the tables needed. This proved to be very time consuming and annoying later on. This table comes from the text file saved in N:\ENGINEERING\permits\Data. This is primarily used for mailing names and addresses
 - CWC Data from Tim Curtin – this is an imported spreadsheet Tim Curtin used to calculate the low 2 quarters of water use. I was able to do this myself within Excel so have abandoned the need for this
 - Queries – there are numerous queries that I used to check to verify that the CWC and LUT were working and I was getting all of the residential properties accounted for.
 - Ultimately the query **_USE THIS QUERY FOR OCT 2021 BILLING** was the final one used to hold all of the data that was then used to prepare the mail-merge document, and the 3767 PDF files for individual residential units. It draws data from **USE THIS TABLE FOR**

BILLING FROM XLS identified in file 1.e above; *9-13-21 from JG FOR Oct2021* from 1.c above, and *parcelTXT 9-13-21* from 1.f.ii above

2. Once the data is provided by CWC, I recommend the following steps in order to first determine the Annual Billing amount
 - a. The first step in the process has to be to determine the revenue needed to manage the collection system and expenditures. This has to be coupled with an understanding of the Sewer Fund undesignated fund balance to properly determine the required revenue.
 - b. Next, we will receive Connecticut Water data – ideally within the first week of January. This data requires significant clean up to make it useable. Using Excel:
 - i. Identify various codes for commercial and residential and standardize to two (res and com)
 - ii. Split Commercial and Residential accounts into separate tables in Excel
 - iii. Create new columns for low 2 quarters summed, and the linking servaddr field and perform the conversions to fill these in
 - iv. Provide commercial table to Tim Foster so that he can manipulate the data and determine EBU's for commercial billing. This will become data needed as part of the entire billing picture
 - v. Link residential table to the LUT previously prepared and perform mismatch queries.
 1. Identify missing links and correct
 2. Identify accounts that have changed and add the PID link
 3. Identify new accounts and add the PID link
 - c. Review the Master Sewer Database and make sure it is up to date by identifying new connections added during 2021 and verifying they are accurate within the database
 - d. Import the “cleaned” CWC data to Access, and using the LUT, join the master sewer database to the CWC data.
 - i. Perform mismatch queries to determine if there are any accounts that are not linking
 1. Clean up tables missing data or not linking
 - ii. Verify well-based sewer accounts
 - e. Identify properties that changed ownership in 2021 by reviewing parcel.txt file from 1-3-22 (actually anytime after 12-31-21 but verify with assessor that data is up to date):
 - i. Import the parcel text file into access (note: when doing this must set up the date in advanced settings at import)
 - ii. Query against the sales date field
 - iii. Link query to CWC via LUT and identify which of the properties that sold in 2021 were on CWC water (this will be important as we establish the amount to be charged to the new home owners that were not responsible for the water use during the entire year)
 - iv. Also identify which properties that sold in 2021 are on wells – this is less important because we will prorate well use charge regardless of water use/ownership
 - v. Verify that the Sewer db and parcel.txt file also agree with the water supply for homes sold in 2021
 - f. Start with the rates used in 2021 for commercial and for residential and determine revenue generated